* NLP algorithm -- Word embedding algorithm

BOW (Bag of Words)
TFIDF (Term frequency * Inverse document frequency)
WORD2VEC

① BOW (Bag of Words)

Text data

[
'small dog',
'cute cute cat',
'cute dog'
]

Bag of words

cat	cute	dog	small
0	0	1	1
1	2	0	0
0	1	1	0

Q What is the difference b/w one hot encoder vs bow?

In one hot encoder we have binary form only 0 or 1
In BOW we can have 2, 3, 4 etc also.

② TFIDF = TF * IDF

~~Sentence~~ $TF = \dfrac{\text{No. of repetition of words in sentence}}{\text{No. of words in sentence}}$

$IDF = \log\left(\dfrac{\text{No. of sentences}}{\text{No. of sentences containing words}}\right)$

Sent 1 → good boy
Sent 2 → good girl
Sent 3 → boy girl good

Words	Frequency
good	3
boy	2
girl	2

TF

	Sent1	Sent2	Sent3
good	$1/2$	$1/2$	$1/3$
boy	$1/2$	0	$1/3$
girl	0	$1/2$	$1/3$

IDF

Words	
good	$\log(3/3) = 0$
boy	$\log(3/2)$
girl	$\log(3/2)$

TF * IDF

	good	boy	girl
Sent 1	0	$\frac{1}{2} \log 3/2$	0
Sent 2	0	0	$\frac{1}{2} \log 3/2$
Sent 3	0	$\frac{1}{3} \log 3/2$	$\frac{1}{3} \log 3/2$

③ Word2vec
Word to Vector

father → girl
father → son

Rome → Italy

We are trying to find out semantic similarity token from the paragraph using help of world2 vec concept.

word2vec will be slow on big data